

Summary of "The EU AI Act: Between Product Safety and Fundamental Rights"

Overview

This paper evaluates the European Union's Artificial Intelligence Act (AI Act) through the lens of technology law. The authors argue that while the AI Act is heavily couched in fundamental rights language, it is structurally designed as a **product safety instrument**. This "legal medley" attempts to combine product safety mechanics with fundamental rights protection, but it risks creating an inadequate framework for both.

The paper frames its critique around three classic "law & tech" dilemmas: **the pacing problem**, **the means-end mismatch**, and **regulatory inheritance**.

1. The AI Act as Risk-Based Regulation

The AI Act rejects a uniform approach, opting instead for a prescriptive, top-down risk segmentation that defines three core legal regimes based on the application context:

- **Unacceptable Risk (Prohibition):** Covers AI practices that violate fundamental EU values (e.g., subliminal manipulation or exploiting vulnerabilities to cause harm). Market access is strictly denied.
- **High Risk (Harmonized Rules):** Covers AI used in safety components of regulated products or applications listed in Annex III. Market access is conditional upon strict technical requirements (ex-ante) and post-market monitoring.
- **Minimal Risk (Residual):** Governs the vast majority of AI systems (estimated 85–95%). No new legal rules are introduced under the Act; they remain bound by baseline regimes like the General Product Safety Directive (GPSD).

The Tiers at a Glance

Risk Level	Definition	Legal Regime	Core Instrument	Other Normative Sources
Unacceptable	Article 5	Prohibition	Precautionary Principle	Member States cannot add/remove applications.

High Risk	Article 6	Harmonized rules at EU level	New Legislative Framework	EU provides norms; Member States cannot legislate.
Minimal Risk	Residual	General safety requirement	General Product Safety Directive	Existing sector law; Member States can add norms.

2. Specifying "AI" Functionalities: The Pacing Problem

The AI Act assumes that lawmakers can foresee and categorize the specific "use cases" or "functionalities" of an AI system to assign its risk tier. The authors highlight how this fails against rapid technological evolution:

- **The Rise of General-Purpose AI (GPAI):** Systems like Large Language Models (LLMs) lack a single, predetermined function and can be fine-tuned for vastly different contexts.
- **The Pacing Problem:** Regulators are caught in a classic quandary. If they legislate early, they lack the data to address unpredictable safety risks (e.g., delegating knowledge production, chatbots of deceased persons). If they wait, adverse real-world impacts may already be irreversible.
- **The Allocation Trade-off:** Determining who bears the compliance burden creates a hard trade-off. Forcing developers to comply may stifle innovation incentives, but developers possess the actual technical capability to mitigate risks—capabilities that end-deployers often lack.

3. Shoehorning Fundamental Rights: The Means-End Mismatch

The core tension of the Act is that it tries to achieve a **fundamental rights protection end** using **product safety means**. This creates a conceptual collision between two completely different legal rationales:

- **Incompatible Conceptions of Risk:** Product safety relies on an *actuarial, probabilistic, discrete, and unidimensional* view of risk, where harms are measurable and often financially compensable. Fundamental rights violations, however, are *socio-technical*,

cumulative, and qualitative (e.g., loss of human dignity or the erosion of societal trust via deepfakes), which cannot be reduced to a simple numerical metric.

- **Satisfaction vs. Optimization:** Product safety operates under a logic of *satisficing*—once a technical baseline is met, the system is legally "safe". Conversely, constitutional fundamental rights demand *optimization* and proportionality, meaning rights must be protected and promoted to the maximum extent possible. Shifting to a product safety lens reduces fundamental rights protection to a check-the-box exercise.

4. Existing Institutions: The Issue of Regulatory Inheritance

To ensure swift enforcement, the AI Act piggybacks on existing European product safety infrastructure (e.g., Standardization Organizations, Conformity Assessment bodies, and Market Surveillance Authorities). This creates structural vulnerabilities:

- **Capabilities Challenges:** Market Surveillance Authorities (MSAs) and Standard-Setting Organizations (SSOs) face severe deficits in both **technical literacy** (AI expertise is highly concentrated in a few lucrative private tech firms) and **legal capabilities** (product safety engineers are not trained to conduct complex constitutional proportionality assessments).
- **Legitimacy Deficits:** Delegating the boundaries of fundamental rights to private or quasi-private technical standardization bodies bypasses democratic processes. It weakens *input and throughput legitimacy* because technical committees exclude broader public and political participation.
- **The Harmonization Trap:** Because the AI Act is a "full harmonization" instrument, Member States are legally blocked from implementing stricter national laws to remedy these fundamental rights loopholes or customize enforcement.

Authors' Structural Recommendations

To prevent sacrificing both product safety and fundamental rights on the altar of legal uniformity, the authors propose:

1. **Narrow the AI Act's Scope:** Strip the fundamental rights optimization mandates out of the horizontal AI Act and focus the regulation strictly on the technical safety baselines of AI systems.
2. **Deploy Ad Hoc Instruments:** Address sector-specific risks to fundamental rights via tailored legal instruments that look at the precise context of application, rather than the product itself.
3. **Rely on International Treaties:** Use dedicated human rights frameworks—such as the Council of Europe's treaty on AI and human rights—to supply deep, uncompromised fundamental rights protections instead of simplifying them into technical standards.